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Mixed Crowd Navigation: Perception, Interaction, Planning, and Control

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Keywords

human–robot interaction, crowd perception and analysis, multiagent systems, data-driven control, human behavior prediction, collective behavior optimization

Abstract

We comprehensively survey mixed crowd navigation, focusing on the integration of robotic agents with volitional crowds (humans or human-driven vehicles) to achieve system-wide benefits. The survey is organized following the perception–interaction–planning–control pipeline, examining four core components: (*a*) perceiving global crowd behavior from local robot observations through nonparticipant and participant observation methods; (*b*) modeling volitional agent responses via rule-based and data-driven interaction frameworks; (*c*) predicting crowd dynamics across microscopic, mesoscopic, and macroscopic scales using both traditional and machine learning approaches; and (*d*) synthesizing control policies that guide crowds toward desired states. We address critical challenges such as complex interaction modeling under partial observability, constrained robotic influence, and the need for multiscale behavioral consistency. Key applications span pedestrian crowd management and mixed traffic control. We also

highlight emerging trends in mixed crowd navigation, including the use of deep reinforcement learning and foundation models, while identifying persistent challenges in human irrationality modeling, compliance prediction, and privacy-preserving algorithms for real-world deployment.

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1. INTRODUCTION

Crowds consist of individual agents with their own dynamics who also exhibit collective behavior through mutual interactions (1). This dual nature has made them an active and essential subject of study across robotics, control theory, traffic engineering, physics, and biology. Research on crowds falls into two broad categories: Engineering focuses on robotic crowds (e.g., drone formations and industrial robot teams), while science investigates emergent patterns in volitional crowds (e.g., pedestrian groups and vehicular traffic).

Robotic and volitional crowds have key differences. First, robotic agents have fully observable states, whereas volitional entities possess partially observable (and noisy) states. Second, robots communicate directly through predefined protocols, while volitional entities exchange information indirectly via verbal and nonverbal cues, introducing delays and ambiguities during interactions. Third, robotic agents are programmed or AI-driven to perform tasks optimally, while volitional entities act based on instinctual responses, often exhibiting suboptimal behaviors influenced by diverse environmental factors.

Over the past decades, advancements in robotics and AI have propelled the widespread adoption of autonomous agents in society. As robots become more prevalent in human environments, mixed robotic–volitional crowds have emerged. For example, self-driving and driver-assisted vehicles now interact frequently with human-driven traffic, while low-cost service robots, such as delivery bots and outdoor cleaners, operate in pedestrian-heavy environments. Many smart city initiatives envision autonomous robots handling diverse tasks, from street cleaning to public security, promising to reduce human workloads while enhancing efficiency and service quality. As these scenarios transition from vision to reality, interactions between robotic and volitional crowds will become increasingly frequent and complex.

Mixed robotic–volitional crowds exhibit more complex and diverse behaviors than homogeneous groups, creating significant challenges for crowd modeling, prediction, and optimization. This complexity raises a critical question: How can robotic agents leverage strategic policies to optimize the behavior of mixed crowds? Consider practical scenarios where a small number of robotic agents must influence predominantly human crowds—self-driving vehicles regulating traffic flows in mixed-autonomy transportation systems, or mobile robots managing pedestrian crowds to prevent congestion while optimizing tasks like environmental monitoring or cleaning efficiency. Successfully addressing these challenges will advance human–robot coexistence so that robots can assist with daily tasks, enhance productivity, and ultimately improve quality of life. To achieve this goal, we need to overcome three fundamental obstacles in mixed crowd navigation:

- Complex interaction modeling: Mixed robotic–human crowds exhibit sophisticated dynamics that transcend simple rule-based models or closed-form equations, demanding advanced computational approaches to capture emergent behaviors and nonlinear interactions.
- Partial observability under uncertainty: Robotic agents must infer real-time crowd states from incomplete and noisy sensor data, making accurate state estimation a critical prerequisite for effective intervention strategies.
- Constrained influence and underactuation: Each robotic agent can directly influence only a limited number of nearby humans, creating inherent underactuation where the control

inputs are significantly fewer than the system's degrees of freedom. This spatial and numerical limitation transforms crowd optimization into a challenging control problem requiring sophisticated centralized or decentralized policies to achieve global objectives—such as reducing crowd density, maintaining orderly flow patterns, or preventing dangerous congestion—through strategically coordinated local interventions.

This survey provides a comprehensive review of progress in mixed crowd applications, identifying key success factors and persistent challenges. We organize our analysis around a perception–interaction–planning–control pipeline that encompasses four core components, as shown in **Figure 1**: (*a*) perceiving global crowd behavior from local robot observations, (*b*) inferring how volitional agents respond to stimuli through both passive observation and active probing, (*c*) predicting volitional crowd dynamics using historical data, and (*d*) synthesizing learning-based feedback policies that guide crowds toward desired states by leveraging spatiotemporal crowd predictions and interaction dynamics models. By presenting a thorough overview of mixed crowds from an algorithmic robotics perspective, we aim to provide valuable insights that will guide future research and enable practical deployment of robotic agents for crowd optimization and interaction.

2. DIFFERENCES WITH OTHER RELATED SURVEYS

Our work addresses mixed crowd navigation through three key contributions. First, we analyze mixed crowds characterized by complex robotic–volitional interactions—a significant gap in existing surveys that focus exclusively on purely volitional crowds (1), robotic systems (2, 3), or simple mixed teams (4, 5). Second, we tackle critical technical challenges that prior surveys have overlooked, including spatiotemporal behavior prediction, inference of volitional behaviors through passive and active sensing, and data-driven control strategies. Third, in contrast to traffic-focused studies (6) that emphasize analytical solutions, we highlight data-driven methodologies that can be applied across diverse crowd types, granularities, and stimuli.

3. TAXONOMY

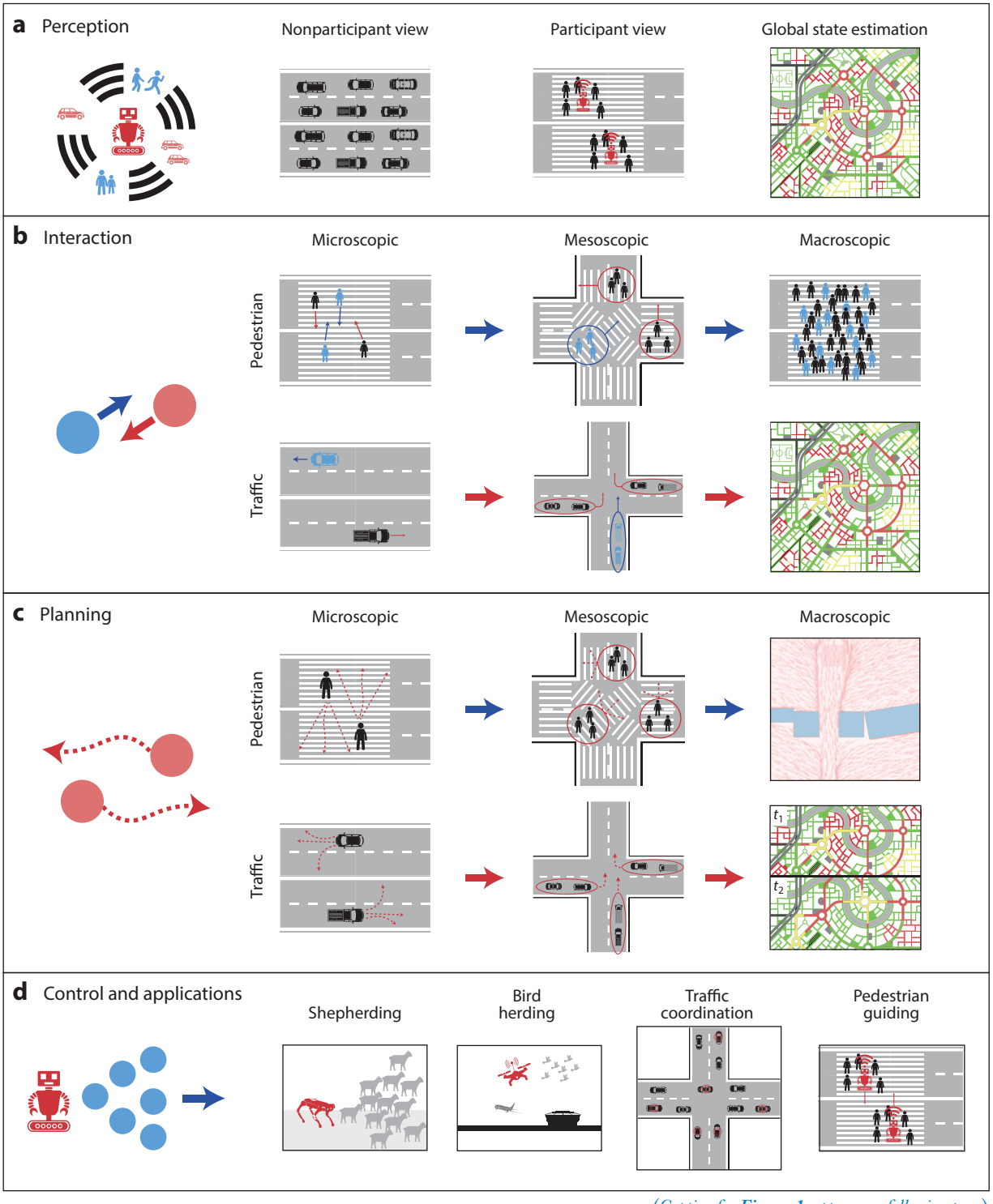
We evaluate the maturity and challenges of mixed crowd studies across four components—perception, interaction, planning, and control—within the optimization pipeline, focusing on robotic regulation of volitional crowds, i.e., human and traffic crowds. Each work is classified along four dimensions: (*a*) the component addressed, (*b*) algorithmic granularity (microscopic, mesoscopic, or macroscopic), (*c*) targeted volitional agents [pedestrians or human-driven vehicles (HVs)], and (*d*) solution design approach (rule-based versus data-driven).

3.1. Key Components

We examine the core components within mixed crowd navigation: perception (inferring occluded agents and crowd flows), interaction (modeling the response of volitional agents to robotic and peer stimuli), planning (predicting multiscale crowd evolution under robotic influence), and control (guiding crowds toward desired densities and behaviors under safety and social constraints).

Effective mixed crowd navigation necessitates tight coupling among these components. Perception informs real-time interaction modeling, which anticipates the nuanced responses of humans and vehicles to guide multiscale planning. Planning synthesizes crowd dynamics into robust trajectories, while control executes actions sensitive to kinematic/social constraints, simultaneously perturbing the crowd. Crucially, feedback closes the loop: Observed reactions (e.g.,

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(*Caption for* ***Figure 1*** *appears on following page*)

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Figure 1 (*Figure appears on preceding page*)

The four aspects of mixed crowd navigation taxonomy: (*a*) perception (estimating global crowd behavior from robot observations), (*b*) interaction (modeling agent responses to stimuli via passive observation and active probing), (*c*) planning (predicting volitional crowd dynamics using historical data), and (*d*) control and applications (executing tasks such as herding, traffic coordination, and pedestrian guiding).

altered pedestrian paths or traffic wave mitigation) refine predictions and replanning. Such synergy allows adaptable challenge mitigation [like predicting HVs' merge delays from turn signals and adjusting robotic vehicles (RVs) to control buffers], balancing fluency, safety, and flow efficiency through integrated perception, interaction modeling, planning, and control.

3.2. Multilevel Formulation

Mixed crowd navigation operates across three scales. At the microscopic level, individual agent behaviors and interactions are governed by unique rules that are simulated in detail. At the macroscopic level, the crowd is treated as a continuous medium, with collective dynamics analyzed through mathematical frameworks that track density, velocity, and other aggregate measures. The mesoscopic level bridges these two perspectives by modeling clusters of agents, tracking group dynamics while accounting for intergroup interactions. The formulation of cross-level interactions is discussed in **Supplemental Appendix 1**.

Supplemental Material >

3.3. Types of Volitional Agents

Volitional crowds differ fundamentally from robotic teams due to their goal-driven resistance, irrational behaviors, and complex social dynamics. Unlike robotic systems, volitional crowds cannot be directly manipulated through physical contact; instead, they rely on noncontact stimuli such as lights, sounds, and motions. This constraint necessitates solutions that combine physics-based reasoning with machine learning. By unifying these technical aspects, our survey highlights the distinct challenges and interdisciplinary strategies required for mixed crowd navigation.

The heterogeneity of pedestrians and HVs fundamentally shapes mixed crowd navigation strategies for mobile robots and RVs. Human crowds require socially aware adaptability, while traffic demands kinematic precision and infrastructure coordination. This necessitates tailored approaches across perception, interaction, planning, and control:

- Perception: Strategies for mixed pedestrian crowds, characterized by slow-moving, omnidirectional agents in unstructured environments, prioritize tracking erratic or occluded individuals. Mixed traffic crowds, in contrast, operate in high-speed settings (e.g., 30 m/s) where vehicles follow nonholonomic constraints and road rules. Hence, long-range sensing and interpretation of signals (e.g., turn indicators) become critical.
- Interaction: When interacting with pedestrian crowds, robots must adhere to implicit social norms by interpreting humans' body language, gaze, and proxemics to subtly influence paths through speed or positional adjustments while minimizing disruption. However, humans' inherent unpredictability amplifies interaction uncertainty. For mixed traffic crowds, explicit protocols govern interactions through rule-bound signals and traffic laws (such as right-of-way hierarchies). The delayed responses of human drivers (1–3 s) also require anticipatory models to synchronize RV maneuvers.
- Planning: Pedestrian crowds, with their microscopic diversity arising from distributed stimuli, require mesoscopic and macroscopic planning approaches to manage emergent stochasticity. Traffic crowds, governed by lane discipline and kinematic chains, enable more deterministic predictions based on structured movement patterns.

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- Control: Pedestrian crowd control prioritizes real-time adaptability, dynamically adjusting to directional shifts and mesoscale social dynamics (such as group cohesion) while balancing proactive goals (such as bottleneck clearance) with safety buffers to accommodate low injury tolerance. Traffic crowd control, however, emphasizes kinematic precision and flow stability by optimizing vehicular sequences to dampen stop-and-go waves and enforce protocols during merges. RV policies must balance stringent dynamic constraints (such as centripetal force limits during lane changes) with varied driver compliance patterns (such as aggressive tailgating versus cautious yielding).

3.4. Rule- or Learning-Based Paradigm

Both rule-based methods (reliant on explicit protocols) and learning-based approaches (data-driven) have been applied to mixed crowd navigation, with method selection depending on environmental structure, safety requirements, and behavioral complexity. Rule-based techniques dominate pedestrian scenarios, where velocity obstacles (7) and social force models (8) enable real-time collision avoidance through geometric constraints. Rule-based approaches also power traffic management such as signal optimization and lane keeping of vehicles, where structured scenarios and clear protocols prioritize safety and interpretability.

Learning-based methods excel at adapting to complex, context-driven behaviors that resist manual rule design. For example, deep reinforcement learning (RL) navigates chaotic crowds by inferring latent social norms (9), while transformers manage intersection flows by capturing multiagent negotiation dynamics (10). These methods also demonstrate superior adaptability to heterogeneous agents with dynamic behaviors, such as crowds undergoing phase transitions (11) or traffic fluctuating between peak and off-peak periods (12).

Granularity also shapes technique choices in mixed crowd navigation. For instance, in the CIRCLES [Congestion Impacts Reduction via Connected Automated Vehicle (CAV)-in-the-Loop Lagrangian Energy Smoothing] project (12), microscopic pedestrian control integrates rule-based constraints (such as personal space buffers) into learned navigation policies, while macroscopic traffic optimization merges physics-inspired flow models with data-driven congestion predictors. These hybrid solutions integrate rule-based safety with learning architectures, balancing regulatory compliance and adaptive flexibility for real-world mixed crowd applications.

4. LITERATURE REVIEW

We structure our review around the key components of mixed crowd navigation and highlight emerging trends and unresolved challenges.

4.1. Perception

There are two approaches in observing crowds: nonparticipant observation and participant observation. Nonparticipant observation offers a top-down perspective, enabling robots to monitor crowd density, track movement trends, or detect large-scale disruptions such as stampedes. While this method minimizes interference with human activities and avoids spatial constraints, it struggles to capture subtle social interactions or localized issues. In addition, its ability to track individual trajectories raises privacy concerns. Participant observation, on the other hand, involves robots immersed in crowds analyzing microlevel dynamics. Equipped with proximity sensors, microphones, and tactile feedback, they can gather rich data on group cohesion, verbal exchanges, and stress signals. However, this approach narrows the robot's field of vision and risks distorting natural crowd behaviors (the observer effect). Moreover, navigating densely packed crowds presents significant technical challenges in dynamic environments.

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Selecting the appropriate observation method is crucial for effective human–robot interaction research. Nonparticipant observation proves most effective for macroscopic monitoring, while participant observation excels in mesoscopic and microscopic studies of human–robot interaction. The choice between these approaches depends on whether the application prioritizes comprehensive situational awareness or detailed behavioral insights, each of which requires careful consideration of ethical implications and technical constraints.

4.1.1. Nonparticipant observation. Nonparticipant observation employs surveillance sensors to capture multiscale crowd features of both pedestrians and vehicles. Macroscopic metrics such as global density, flux, and velocity are then computed via crowd counting.

Recent work has emphasized crowd localization via point-level annotations: Detection-based approaches apply Faster R-CNN (faster region-based convolutional neural network) with nonmaximum suppression (13), map-based methods use Gaussian kernels for head localization (14), and regression-based techniques such as CLTR (crowd localization transformer) (15) employ transformers for direct location mapping. In contrast to macroscopic dynamics, microscopic surveillance involves localizing individuals via optical signals (color/depth) followed by trajectory tracking. Color footage with markers enables behavior analysis (16), while overhead depth sensors operate in real-world settings (17). Recent computer vision advances measure finer individual features—gestures, luggage, body size, and age—that are critical to dynamics. For instance, body rotation extracted from video/depth data predicts macroscopic chiral oscillations in dense crowds (11). High-level traits estimated from video feeds improve real-time behavior prediction (18). Beyond visual sensors, mobile Bluetooth/Wi-Fi signals reveal intermittent motion patterns at events (19) or reconstruct pedestrian routes via MAC address matching (20).

Vehicle monitoring differs significantly from pedestrian crowd observation due to fundamental differences in movement patterns and infrastructure requirements. Roadside monitoring devices, including inductive loops, cameras, and GPS systems, can effectively capture vehicle trajectories with relatively sparse placement. For example, the NGSIM (Next Generation Simulation) project (21) deployed five to eight rooftop cameras to track vehicles on 500-m multilane roadways at 0.1 Hz, while Tennessee's I-24 MOTION (Interstate 24 Mobility Technology Interstate Observation Network) initiative (22) uses 294 ultra-HD cameras to monitor a four-mile interstate section, extracting speed, acceleration, and interaction data.

4.1.2. Participant observation. Recent advances in autonomous robot navigation and decision-making have enabled participant observation studies that deploy robotic agents as mobile sensors to gather detailed spatiotemporal data on crowd dynamics. However, robotic sensors that operate from a first-person perspective, creating partial observability due to visual occlusions and limited sensing ranges, restrict comprehensive environmental awareness. To address this challenge, researchers resort to the bird's-eye view (BEV) method, which generates unified top-down representations that maintain microscopic interaction details.

Pedestrian crowd analysis employs BEV to project human locations into geometrically consistent spatial frameworks (23). This technique mitigates perspective distortions from ground-level cameras, enabling robust spatial analysis in occluded or dense scenes. For instance, Dai et al. (24) employed a multibranch convolutional network to jointly predict head and foot density distributions in the BEV space, and Dendorfer et al. (25) integrated trajectory prediction with BEV coordinates in a multiobject tracking framework.

Traffic applications also use BEV in critical tasks such as localization, obstacle detection, and path planning. BEV can capture semantically rich spatial information by leveraging ground-plane keypoints, multiview imagery, or geometric transformations for downstream tasks. Two approaches are commonly used: bottom-up strategies, which reconstruct scenes from sensor data

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or monocular cameras (26), and top-down approaches, which map high-level semantic features into the BEV space (27).

Despite BEV's potential, challenges persist, particularly occlusion and heterogeneous interactions in dense crowds. While mixed traffic and autonomous driving research mitigates these issues via spatiotemporal aggregation of multiview data, adapting such solutions to dynamic human crowds remains unresolved. Beyond robotic perception, participant observation has incorporated other methods, such as leveraging human mobility data and social media traces for abnormality detection (28). However, participant observation risks altering crowd dynamics and influencing volitional agents' behavior. Mitigating or exploiting this issue requires aligning perception with interaction, which we discuss in Section 4.2.

4.1.3. Crowd global state estimation. Neither nonparticipant nor participant observation fully captures crowd dynamics. Fixed sensors (e.g., drones and cameras) offer limited spatial coverage (e.g., road segments or urban blocks), while onboard sensors detect localized conditions in their vicinity. These partial views, combined with incomplete prior knowledge of crowd behaviors, hinder accurate estimation of macroscopic states (density, speed, and flow) across large-scale areas. However, a precise global estimate remains essential for optimizing mixed crowd flows, as planning and control strategies rely on crowd-wide insights.

Pedestrian crowds exhibit high motion diversity and unpredictability due to their reduced reliance on environmental geometries, making it challenging to accurately estimate global states. Solutions include aligning sensor observations using intensity patterns to enable consistent crowd tracking in cluttered environments (29). Data from pedestrian-mounted sensors or robotic agents within the crowd can enhance this alignment. For instance, aggregating sparse GPS data from city-scale event visitors can enable pedestrian map construction (30). Data aggregation can also involve mapping algorithms that fuse a robot's local crowd perceptions to model macroscopic crowd behaviors (31). Map uncertainty can be greatly reduced via optimal robot navigation in crowded environments by leveraging prior knowledge such as obstacle geometry (32) or conservation laws (33).

Regarding traffic, conventional global state estimation leverages sparse partial observations to calibrate traffic models. This approach includes partial differential equations (PDEs) for traffic dynamics (34), generative models [e.g., generative adversarial networks (GANs) and diffusion models] for behavior prediction (35), and conservation laws to enforce vehicle continuity across spatiotemporal perceptions (36). Estimation accuracy is further enhanced by incorporating crowdsourcing data such as taxicab trajectories (37) and mobility patterns (28). Advanced techniques, including compressed sensing (38, 39), metamodel-based methods (40), diffusion convolutional recurrent neural networks (RNNs) (41), graph CNNs (42), and imitation learning (IL) (43), have also been adopted to reconstruct dense traffic states from sparse observations. Recent work such as the CIRCLES project (12) improves spatiotemporal resolution by forecasting congestion frontiers (e.g., bottlenecks and shockwaves) via neural networks and fusing these predictions with real-time microscopic data from RVs (GPS, speed, and local conditions) to address latency in macroscopic nonparticipant observations.

4.1.4. Key takeaways. The key takeaways from this section are as follows:

- Mixed crowd perception uses nonparticipant observation for broad monitoring and participant observation for detailed analysis.
- No single method captures complete crowd dynamics, making large-scale state estimation difficult with limited sensor coverage.
- BEV perception creates unified overhead views that capture fine details while avoiding ground-level spatial constraints.

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- Pedestrian tracking requires dense sensors for unpredictable movement, while vehicle monitoring uses sparse sensors on predictable roadways.

4.2. Interaction

Understanding crowd dynamics requires multiscale modeling that links individual agent behaviors with emergent collective phenomena. At the microscopic level, interactions determine how agents react to stimuli such as nearby individuals, robots, or environmental constraints. These fine-grained interactions collectively shape group behaviors and macroscopic phenomena, including stop-and-go waves in unidirectional flows (44), self-organized vortices in mosh pits (45), or spontaneous rotational movements during "crowd quakes" (11). While macroscopic dynamics adhere to principles such as momentum conservation (similar to fluid mechanics), microscopic models must account for physical, social, and environmental factors to explain emergent complexity. Traditional rule-based methods, which rely on predefined formulas and manual tuning, often fail to capture the nonlinear heterogeneity of real-world interactions, especially in high-density scenarios where crowded spaces risk fatalities. Data-driven approaches use observed behaviors to approximate true interaction patterns, avoiding rigid assumptions. Evaluating these methods depends on their alignment with real-world observations: Microscopic models must match empirical trajectories, while macroscopic methods must ensure consistency with conservation laws.

4.2.1. Rule-based interaction model. For mixed pedestrian–robot crowds, social force models are widely adopted as the microscopic framework, simulating interactions via repulsive and attractive forces (8). These models replicate natural crowd behaviors in normal and emergency scenarios and integrate robots through artificial forces: either repulsive dynamic obstacles to enhance evacuation efficiency (46) or attraction-based leaders for group guidance (47). Hybrid approaches merge social forces with panic propagation models for exit selection (48) or optimize robot controllers by formalizing human–robot force dynamics (49). While effective for small groups, these methods face significant scalability challenges when applied to large mixed crowds of robots and humans.

Macroscopic models describe crowd dynamics via PDEs governing density and velocity (50). However, embedding robotic influence into continuum-based formulations remains challenging due to their focus on group-level dynamics. Hybrid or mesoscopic models bridge this gap by ensuring equivalence between microscopic interactions (e.g., social forces) and macroscopic crowd behavior (51). These methods link individual actions to collective density and flow patterns (52), leveraging real-time macroscopic data (Section 4.1) to guide robots in influencing local agents and global flows. Initially applied to robotic swarms, hybrid frameworks can further address mixed crowd evacuation (53), combining microscopic social forces with hydrodynamic equations.

For mixed traffic, rule sets have been used to govern HVs and RVs to account for behavioral differences. Microscopic HV interaction models based on ordinary differential equations (ODEs) focus on longitudinal dynamics, simplifying lateral maneuvers such as lane changing. Follow-the-leader models (54) assume that each vehicle adjusts its speed based on the preceding vehicle's distance and velocity, aligning with continuum models such as the Lighthill–Whitham–Richards model at high densities. However, follow-the-leader's deterministic nature risks unphysical outcomes, such as negative velocities. The optimal velocity model (55) addresses delayed driver responses to replicate stop-and-go waves, while the intelligent driver model (56) balances free-flow speed objectives with safe gap maintenance via acceleration equations. RVs, in contrast, employ rule-based frameworks that leverage advanced sensing and communication. Adaptive cruise control dynamically adjusts speed to maintain safe headways (57). Cellular automata models are used to discretize space and time, applying localized rules for acceleration and lane changes (58).

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Collision avoidance in high-speed traffic demands more precise modeling beyond pedestrian crowds. The Gipps model (59) enforces safety by constraining speed under sudden leader braking, though homogeneity assumptions limit realism (60). Extensions address perception delays (61), while stochastic models such as Langevin-based noise (62) and Gaussian mixtures (63) enhance behavioral diversity.

At the macroscopic level, rule-based models analyze system-wide traffic flow through density, speed, and flow. Multilane gas-kinetic frameworks simulate adaptive cruise control behaviors with lane change dynamics (64), while discrete second-order models resolve heterogeneity by segregating HV and RV dynamics (65). Despite recent progress, studies remain scarce because collective behaviors are typically examined at the macroscopic level. Significant challenges persist in capturing how behaviors couple and interact across different scales.

4.2.2. Data-driven interaction modeling. Data-driven modeling of mixed pedestrian crowds has relied on passively collected datasets containing annotated positions, trajectories, and group behaviors from visual recordings. Earlier work includes Social LSTM (social long short-term memory) (66), Social-GAN (67), and multimodal frameworks such as DESIRE [deep stochastic inverse optimal control (IOC) RNN encoder-decoder] (68) and Behavior-CNN (69). Recent studies have adopted attention-based (70) and graph-based paradigms, examples of which include SoPhie (71) and SFEM-GCN (social force embedded mixed graph convolutional network) (72).

RL complements supervised learning in modeling interactions through rewards. Early RL methods use manually designed rewards to train behaviors such as collision avoidance (73). Recent advances model heterogeneous interactions (74) and capture nuanced decision-making through adaptive reward structures (75).

Despite these advances, passive data collection is inherently limited in capturing interactions shaped by external stimuli or proactive agents, which is essential for mixed pedestrian–robot crowds. Fortunately, robots can pursue active probing and deliberately influence crowds to uncover latent interaction patterns (9). Beyond active probing, active cooperation complements interaction modeling by enhancing behavioral realism and task efficiency. For example, Kretzschmar et al. (76) applied inverse RL to infer cooperative navigation behaviors from human trajectories. Trautman et al. (77) modeled collision avoidance as a joint task for cooperative human–robot trajectory modeling.

Data-driven modeling of traffic leverages machine learning to infer interaction patterns between HVs and RVs. Occupancy-based methods that encode scene dynamics into probabilistic BEV grids have emerged as a prominent approach (78). Further developments unify spatial occupancy probabilities with motion dynamics through occupancy flow fields (10). While occupancy-based methods excel at microscopic spatial interaction encoding, they struggle with temporal dependencies and computational demands at scale. Trajectory-based methods address these limitations by explicitly modeling individual behaviors with historical motion and environmental context (79).

To address the inherent uncertainty in human driving behaviors, multimodal prediction methods model multiple plausible future trajectories in mixed traffic (80). Graph-based reasoning further advances joint prediction by encoding relational dynamics (81). Autoregressive approaches address temporal dependencies by modeling interactions through stepwise prediction (82). However, autoregressive frameworks suffer from error accumulation over long prediction horizons. Conditional prediction methods address this issue by decomposing joint distributions into causal factors, achieving computational efficiency through explicit traffic interaction modeling (83).

Existing interaction models operate independently from planning, predicting agent responses without considering ego vehicle intentions. Integrated prediction–planning frameworks address

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this by modeling bidirectional ego–agent influences, creating closed-loop negotiation systems. Chen et al. (84) pioneered this integration through tree-structured policy planning. Another key limitation of existing vehicle interaction models lies in their reliance on passive observation, which restricts their effectiveness for RVs operating among HVs with uncertain latent states (e.g., attentiveness or aggressiveness). Active interaction methods mitigate this issue by letting RVs strategically probe these hidden states through deliberate maneuvers. Sadigh et al. (85) used probing actions to clarify human states such as attentiveness and driving style. Hu et al. (86) balanced probing and task performance with runtime safety guarantees. However, these methods remain constrained to low-speed environments due to safety risks in high-speed traffic.

4.2.3. Key takeaways. The key takeaways from this section are as follows:

- Mixed crowd interaction modeling requires bridging microscopic individual behaviors with macroscopic collective phenomena.
- Rule-based models are interpretable but fail to capture the nonlinear, heterogeneous interactions in dense crowds, prompting a shift to data-driven methods that better approximate real-world dynamics.
- Active interaction methods let robots probe latent human states through strategic maneuvers, overcoming the limits of passive observation in contexts with uncertain human intent.

4.3. Planning

Mixed crowd planning models the temporal evolution of autonomous crowds by integrating interaction dynamics (Section 4.2) with historical states, robotic stimuli, and environmental constraints. A core challenge lies in ensuring behavioral consistency across microscopic (individual), mesoscopic (group), and macroscopic (crowd) scales. Key complexities include reconciling rule-based methods (e.g., social forces) with data-driven approaches (e.g., RL) for robustness under partial observability, modeling context-dependent decisions like panic-induced path deviations, and addressing scenarios from routine operations to emergencies. Addressing these complexities demands synergizing rule-based and data-driven planning, both of which can benefit from sophisticated crowd simulation tools (87, 88). Rule-based methods ensure safety, collision avoidance, and regulatory compliance; data-driven approaches capture emergent behaviors and context adaptation under partial observability. Hybrid frameworks embed rule-based constraints into learning while using data to tune rules or predict irrational behaviors, enabling scale-consistent hierarchical planning.

4.3.1. Rule-based planning. Rule-based mixed pedestrian–robot crowds evolve in unstructured scenes characterized by spontaneous path deviations, socially driven interactions, and context-dependent decisions influenced by factors such as comfort, culture, and emotion.

Realistic pedestrian crowd planning incorporates diverse psychological and physiological factors. Personality models [e.g., OCEAN (openness, conscientiousness, extraversion, agreeableness, and neuroticism) (89)], alongside emotion frameworks such as OCC (Ortonny, Clore, and Collins) (89), influence decision-making and behavioral responses. Appraisal theory and stress models describe agents' reactions to environmental stimuli. Physiological traits and social roles, along with environmental constraints such as exit widths and evacuation bottlenecks, are crucial in shaping crowd dynamics (90). Mesoscopic behaviors, informed by social comparison theory and collision-aware grouping strategies, include leader–follower dynamics and responses to pandemics (91). While macroscopic planning methods exist, they are less prevalent in pedestrian

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crowd modeling compared with traffic systems, largely due to the emphasis on individual- and group-level dynamics in open environments.

In contrast to pedestrian planning, traffic planning predicts mixed traffic evolution using domain-specific constraints (e.g., road geometry and vehicle dynamics). Microscopic methods focus on individual vehicle trajectories using kinematic or dynamic constraints (92), relying on physics-based assumptions effective in structured environments. Mesoscopic models, using vehicle clustering or gas-kinetic analogs, face challenges in application due to complexity (93). Macroscopic approaches like the Lighthill–Whitham–Richards and Aw–Rascle–Zhang models simulate traffic flow but are limited by nonequilibrium phenomena and scenario suitability. Hybrid models combining Aw–Rascle–Zhang with discrete vehicle simulations address multilane dynamics but increase computational demands (94).

4.3.2. Data-driven planning. Recent advances in pedestrian crowd planning leverage machine learning to bypass manual parameter tuning. RL trains agents to navigate complex environments through trial and error, while data-driven models trained on human trajectories generate crowds mimicking real-world behavior (75). Early domain-aware approaches relied on labor-intensive tools like cage-constrained motion (95), which lacked environmental adaptability. Subsequent techniques improved realism by learning motion styles (90) and emergency responses (96), although they still required predefined trajectories. Ji et al. (97) addressed these gaps with a text-guided system that converts natural language prompts into structured instructions via large language models such as GPT-4. Two trained diffusion models, one predicting agent start–goal distributions and another generating semantic-aware velocity fields, guide collision-free navigation using reciprocal velocity obstacle simulations (7), with dynamic velocity adjustments to prevent overcrowding. Trained in synthetic randomized environments and paths, the system generalizes to unseen scenarios, operates in real time, and produces high-quality crowd trajectories. Unlike agent-specific latent diffusion models for short-term predictions (98), the method generates global velocity fields in a single inference pass, enabling efficient large-scale planning without sacrificing realism.

As discussed in Section 4.3.1, rule-based traffic planning methods rely on mathematical or physical equations, limiting their ability to incorporate realistic driving behaviors. In contrast, data-driven traffic planning has emerged as a prominent alternative, leveraging real-world historical data to predict realistic future behaviors in mixed traffic environments (99). The microscopic traffic planning methods, which are broadly used in autonomous driving, can be categorized primarily as RL or IL frameworks, while IL can be subdivided into behavior cloning, inverse RL, and model-based IL. **Supplemental Appendix 2** describes the specific datasets and network backbones used across these methods.

Supplemental Material >

Behavior cloning frames IL as a supervised task, mapping expert demonstration states to action distributions. Recent advances address this via transformers tokenizing actions, states, or controls with variable-length trajectory histories (100). However, behavior cloning struggles with covariate shift, where the learner's policy-induced states deviate from the expert's distribution (101). Guo et al. (43) introduced online augmentation using a variational autoencoder that jointly reconstructs expert and learner state distributions, aligning augmented expert states with the learner's distribution while preserving proximity to original data. This bridges the covariate shift gap without compromising trajectory realism.

Inverse RL addresses covariate shift by inferring reward functions from expert demonstrations and deriving policies through online RL. By enabling agents to interact with the environment during training, inverse RL mitigates state distribution mismatches between learner and expert. Inverse RL variants such as generative adversarial IL (GAIL) (102) use adversarial training to align

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learner-generated trajectories with expert data. While GAIL resolves covariate shift in single-agent settings, its effectiveness diminishes in multiagent scenarios due to low sample efficiency and dynamic complexity. To mitigate these issues, GAIL-based methods also exhibit unrealistic behaviors, including collisions and abrupt maneuvers, and lack interpretable reward reconstruction. In contrast, adversarial inverse RL (103) explicitly models explainable reward functions, enabling policy transfer to novel scenarios, as demonstrated in intersection dynamics modeling (104).

Model-based IL addresses covariate shift by minimizing the discrepancy between learned trajectories rolled out in a differentiable environment model and expert trajectories via backpropagation. TrafficSim (105), for instance, models multiagent joint policies using a variational autoencoder and propagates gradients through time-stepped simulations via the reparameterization trick. But such closed-loop methods often suffer from gradient explosion in long-horizon rollouts (106). MRIC (model-based reinforcement–imitation learning framework with temporally abstracted mixture-of-codebooks) (106) stabilizes training with open-loop regularization, keeping policies close to expert distributions. While most model-based IL methods minimize hand-crafted metrics (e.g., Huber loss or Kullback–Leibler divergence), RTR (reinforcing traffic rules) (107) integrates adversarial training through model-based GAIL, where a discriminator measures trajectory similarity, and gradients from the environment model refine policies via closed-loop optimization, balancing realism with training stability.

Recent RL-based planning advances highlight the benefits of jointly optimizing decision-making and trajectory generation, as deep RL methods can flexibly adapt lane changes in dynamic highway traffic, outperforming modular pipelines in safety and efficiency (108). Interaction-aware RL policies further guide optimization-based planners to proactively negotiate dense or uncertain traffic, reducing collisions and increasing success relative to purely learning-based or optimization-only baselines (109). Hybrid hierarchical frameworks—combining high-level RL with classical or sampling-based motion planners—enable robust behaviors in complex real and simulated multiagent scenarios, and cyclic behaviors have been shown to be crucial for bridging the sim-to-real gap (110). In longitudinal planning, RL-based approaches such as ADRL-MTF (Adam optimization-based deep reinforcement learning model for mixed traffic flow control) (111) achieve significant fuel savings and real-time performance across varied traffic conditions, supporting practical deployment.

The mesoscopic traffic planning approach involves clustering and processing traffic participants into distinct groups, rather than directly predicting individual trajectories or the overall traffic volume of the entire network (112). Macroscopic traffic planning methods leverage various advanced techniques to forecast traffic flow accurately. Recently, convolutional networks of spatiotemporal graphs have emerged (113), integrating spatial and temporal dynamics through deep learning based on graphs, which allows for more comprehensive traffic flow predictions that consider the interconnected nature of road networks. These methods collectively advance the field by addressing both the temporal and spatial complexities of traffic flow forecasting.

4.3.3. Key takeaways. The key takeaways from this section are as follows:

- Mixed crowd planning integrates interaction dynamics, historical states, and environmental constraints to ensure consistency across microscopic, mesoscopic, and macroscopic scales.
- Rule-based planning methods struggle with complex, nonlinear behaviors despite addressing psychological and social diversity.
- Data-driven crowd planning approaches improve accuracy by leveraging real-world data and capturing spatiotemporal dependencies.
- Traffic planning benefits from structured environments but requires different approaches at different scales.

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4.4. Control

A central challenge in mixed crowd navigation lies in designing control policies that optimize collective behavior. Straightforward approaches rank control actions such as robotic agent motions based on their predicted impact on future crowd dynamics (Section 4.3), leveraging perceived crowd states (Section 4.1) and estimated interaction models (Section 4.2). Key applications include optimizing traffic flow in scenarios such as single-lane roads, intersections, and multilane networks, addressing emergent behaviors such as platooning, speed harmonization, and vehicle spacing (5). Beyond traffic, mixed crowd control can also regulate swarms of living agents across a broad range of biological, agricultural, and urban contexts—for example, by diverting bird flocks from airports (114), herding sheep (115), or enforcing pedestrian social distancing (116).

4.4.1. Rule-based control. By accounting for human behavior and environmental uncertainties during emergencies, rule-based control has been extended to managing evacuation challenges in mixed pedestrian crowds. While individuals often follow signage, protocols, or directives from authorities, panic can disrupt adherence, as is evident in scenarios involving smoke-obscured exits or suboptimal human decisions. Advances in robot–crowd interaction and decentralized navigation enable mixed crowd navigation systems, where robots dynamically regulate pedestrian flows in dense settings to improve evacuation outcomes. As outlined in **Supplemental Appendix 3**, these solutions employ two strategies: (*a*) explicit control through direct commands or physical interventions and (*b*) implicit control via environmental shaping or behavioral nudges, both of which are aimed at coordinating human–robot interactions in chaotic conditions.

Supplemental Material >

Explicit control strategies deploy robots as active navigational aids that provide direct evacuation guidance to human groups. These systems prioritize proactive intervention over reactive responses. For example, Zhang & Guo (117) demonstrated a multirobot leader–follower system that dynamically adapts to crisis scenarios by coordinating human movement through hierarchical robotic structures. To alleviate congestion, Boukas et al. (118) proposed the use of autonomous robots to redirect evacuees from crowded exits to underused alternatives with a cellular automaton model, reducing bottlenecks and evacuation times. Expanding beyond ground robots, Sakour & Hu (119) proposed using unmanned aerial vehicles to monitor crowds via agent-based simulations, transmitting real-time warnings to steer individuals away from hazards. These methods underscore the versatility of explicit robotic control in balancing directive authority with adaptability during emergencies.

Implicit control strategies deploy robots as dynamic barriers or shepherds that subtly shape pedestrian movement through indirect behavioral nudges rather than explicit instructions. They draw inspiration from natural flock guidance, where subtle cues optimize crowd flow. At hubs such as stations and event venues, intersecting pedestrian flows often create congestion, heightening accident risks. Eldridge & Maciejewski (120) optimized robot behavior via genetic algorithms to generate cohesive forces that attract human groups. Yamamoto & Okada (121) introduced periodic robot movements at flow intersections, enhancing pedestrian velocity through localized collision avoidance. Extending this to corridors, Jiang et al. (122) employed adaptive dynamic programming for real-time robot trajectory replanning, optimizing unidirectional outflow. Inspired by animal herding, Zhang et al. (123) developed a distributed algorithm that enables robots to steer crowds using local information, with "outmost push" motions guiding pedestrians cohesively. Multirobot caging systems enhance this approach, encircling crowds via edge-following behaviors and compacting groups to channel evacuees toward exits (124), balancing subtle guidance with structural efficiency.

Both explicit and implicit control strategies illustrate robotics' transformative potential in emergency response. Explicit methods employ robots as direct guides, while implicit strategies

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subtly shape crowd dynamics. Together, these approaches enhance evacuation efficiency, safety, and human–robot interaction models. Future research should refine adaptability across diverse emergency scenarios, with practical implementations in transportation hubs and public venues.

Mixed traffic control employs analytical control laws based on real-time traffic states to optimize throughput, stability, and safety while managing heterogeneity between HVs and RVs. This approach operates at two levels: individual vehicle coordination and traffic system management, each of which targets distinct operational scales. At the vehicle or microscopic level, rule-based mixed traffic control encompasses conventional control algorithms (longitudinal, lateral, lane change, and merging coordination); for example, game-theoretic approaches model negotiation dynamics in mixed traffic (125). Macroscopic traffic control optimizes network performance using interventions such as ramp metering and variable speed limits, with connected vehicles enhancing effectiveness via vehicle-to-vehicle or vehicle-to-infrastructure communication (126).

4.4.2. Data-driven control. The application of robotic control in mixed pedestrian crowds remains cautious due to ethical and safety concerns, particularly the absence of guarantees that learned policies will reliably avoid harm. Recent efforts have focused on multirobot systems that encourage compliance with behavioral norms, such as social distancing during COVID-19, by leveraging persuasive interactions (116). However, ethical debates persist about the appropriateness of robotic influence over human behavior. For nonhuman species, learning-based herding methods—including supervised learning from expert demonstrations and RL—have been explored in simulation (127), but real-world implementations are rare. For example, studies with drones show that parameters like speed, altitude, and auditory cues affect sheep stress responses and herding efficiency (128), while other studies demonstrate drones redirecting bird flocks near airports by optimizing positional influence on flight paths (114). Similarly, Boston Dynamics' Spot robot has been tested as a low-cost herding agent, mimicking sheepdog tactics such as maintaining flight distances, aligning movements parallel to flocks, and simulating indirect eye contact through orientation adjustments (115). Hybrid human–animal–robot collaboration, such as pairing Spot with a live sheepdog to block escapes and drive flocks into pens, has also proven feasible. Despite these advances, scaling data-driven crowd control to real-world agricultural settings or bird–aircraft collision prevention remains challenging, requiring further refinement of safety, reliability, and ethical frameworks.

Data-driven traffic control has seen transformative advancements through learning-based methods, enabling RVs and CAVs to collaborate with HVs for improved traffic efficiency. Unlike classical model-based approaches, modern techniques, including optimization, deep RL, IL, and neural networks, derive control policies directly from data and/or simulations, bypassing reliance on predefined analytical models. Optimization-based methods address mixed traffic control by framing it as a mathematical problem, coordinating vehicles in scenarios such as unsignalized intersections (129). However, their reliance on explicit traffic flow modeling often restricts their capacity to handle the intricate dynamics of mixed human–robot interactions. This limitation has spurred the adoption of RL, which learns adaptive policies through direct interaction, circumventing rigid assumptions inherent to analytical models.

RL has enabled landmark advances in traffic control, such as employing RVs as mobile actuators to dampen traffic oscillations (130). Wu et al. (131) trained deep RL agents using local observations to suppress stop-and-go waves, outperforming traditional controllers. Recent work extends RL to diverse scenarios, including figure-eight circuits (132), highway bottlenecks and merges (133), and intersections (134, 135). However, these studies often simplify real-world complexity by focusing on isolated vehicle interactions rather than system-wide traffic dynamics. To address this, RL has

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advanced into complex freeway scenarios to address system-wide dynamics. For example, Chen et al. (136) demonstrated multiagent RL for merging interactions between on-ramp and mainline traffic.

Data-driven mixed traffic control has also advanced autonomous intersection management, where CAVs negotiate crossings without traffic signals. Simulations demonstrate that such systems achieve near-optimal throughput with only 33% CAV penetration (134) while reducing crash rates, travel times, and emissions (137). Further innovations employ graph neural networks and attention mechanisms to model spatial dependencies and interagent interactions, enabling adaptive control that accounts for vehicle heterogeneity and behavioral uncertainty (138). Beyond single intersections, scalable solutions include dynamic vehicle routing with privacy-preserving crowdsourcing to optimize urban traffic through RVs (139), as well as multiagent RL that generalizes across unseen intersection topologies via multitask and zero-shot transfer learning (140). These studies highlight the potential of learning-based methods to address both localized and city-scale traffic challenges.

Hierarchical architectures are increasingly adopted in mixed traffic control. For example, in freeway settings, hierarchical multiagent frameworks model objectives such as energy management, braking, and trajectory tracking as cooperative agents under game-theoretic structures (141). For dense traffic, Cai et al. (142) introduced interaction-aware hierarchical representations that streamline human–robot interactions across control layers, adapting to varying CAV penetration rates and driving styles. At the urban level, bidirectional hierarchies coordinate infrastructure and RVs. For instance, Sun et al. (143) proposed a system that positions traffic lights as high-level managers that direct regional goals while decentralized RV agents handle local rerouting. Hybrid architectures further merge high-level soft actor–critic policies with low-level model predictive control to enable collision-free maneuvers at unsignalized intersections (144). Collectively, these methods underscore hierarchical control's role in harmonizing scalability, safety, and adaptability across mixed traffic control ecosystems.

As a complement to existing AI methods, deriving human-like control policies from demonstration data could enable interpretable control in mixed traffic. For example, hybrid IL–probabilistic sampling combines precise interaction modeling in close proximity to probabilistic sampling for larger headways, enhancing safety and efficiency across CAV penetration rates (145). Foundation models such as large language models have also been adopted; for instance, ChatGPT can assist nonexperts in designing state spaces and rewards (146) for mixed traffic control.

Recently, the CIRCLES project (12) addressed the critical question of whether data-driven mixed traffic control can yield real-world benefits beyond simulation. By deploying a framework to manage 100 CAVs in live traffic via Lagrangian variable speed limits, the project bridges theoretical scalability and practical implementation. Its dual-layer architecture combines a centralized speed planner, which generates target speed profiles using a high-resolution spatiotemporal traffic state estimated from fused global and vehicle-level perception data, with decentralized controllers for execution. Two core strategies underpin the framework: (*a*) kernel smoothing to spatially average speeds and dampen shockwave oscillations and (*b*) an RL-based space buffer design that creates controlled deceleration zones upstream of bottlenecks. The RL component employs an actor–critic algorithm trained on coupled PDE–ODE traffic models to dynamically optimize space buffer parameters for throughput. Decentralized controllers adapt these speed profiles locally using embedded RL policies, balancing traffic flow, safety, and fuel efficiency. Field tests achieved an 8% reduction in overall traffic density, with localized improvements of 7% upstream, 10% downstream, and 52% during bottleneck congestion. These results demonstrate, for the first time at scale, that learning-based control of RVs can enhance mixed traffic performance in the real world.

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Notably, mixed traffic control need not rely on level 4 (L4)–capable RVs. Instead, it can operate as a recommendation system for L1–L3 systems, guiding drivers to adopt behaviors that benefit overall traffic, such as smooth acceleration, moderate speeds, fuel-efficient routing, or sacrificing waiting time to minimize congestion. While drivers retain control and can override suggestions, this approach leverages the broader L1–L3 market to achieve wide-scale impact. Challenges persist, however, in ensuring driver compliance and maintaining system robustness when recommendations are ignored (140), underscoring the need for adaptive, human-centered designs.

4.4.3. Key takeaways. The key takeaways from this section are as follows:

- Designing robot control policies to strategically influence human crowds to optimize collective behavior is the key challenge of mixed crowd control.
- Two strategies exist: explicit control, which is exerted through direct commands or physical interventions, and implicit control, which is achieved via environmental shaping or behavioral nudges.
- Applications of mixed crowd navigation include optimizing traffic flow, regulating swarms of living animals, enforcing pedestrian social distancing, and more.

5. CONCLUSIONS: APPLICATIONS, GENERAL TRENDS, AND OPEN CHALLENGES

The integration of robotic agents into urban systems is accelerating, reshaping navigation, work, and urban coexistence. In transportation, the growing adoption of L2–L4 autonomous RVs enables practical implementation of eco-driving principles, speed profile optimization, and route selection (147). In public safety domains, autonomous drones enable crowd management at events such as hajj through real-time density tracking and dynamic guidance. Cities such as Shenzhen deploy robots to optimize pedestrian flows during pandemics or peak hours, combining enforcement with adaptive response (148).

All of these innovations are stepping stones toward a future where cities evolve into mixed human–robot ecosystems. Humans contribute creativity; robots enable scalable coordination, transforming fragmented systems into cohesive networks. Designed to prioritize equity and sustainability, robots mitigate limitations of self-interest inherent to humans. Integrating them as collaborators, not competitors, offers a path to transcend the local minima of inefficiency, steering society toward globally optimal outcomes. The goal is clear: leveraging robotic tools to orchestrate complexity, creating smarter, safer, and more resilient communities where both humans and machines thrive.

Nevertheless, the successful deployment of mixed crowd control in urban environments faces three key challenges, as illustrated in **Figure 2**. First, interaction modeling and planning often assume that humans act (approximately) rationally. However, human behavior frequently deviates from rationality due to biases such as hyperbolic discounting, loss aversion, and the peak–end rule (149). Developing interaction models that account for these irrationalities is critical to minimizing the sim-to-real gap in mixed crowd control. Second, robotic agents in mixed crowds can only indirectly influence nearby humans. For example, in mixed traffic control, L4 RVs may adjust their speed or lane position to guide HVs, while L1–L3 RVs provide suggestions (e.g., recommending less congested routes or smoother driving styles). However, these strategies rely on full human compliance, which is often unrealistic: Experienced drivers may circumvent robotic constraints through aggressive maneuvers, while others might prioritize personal goals like shorter travel times over system-wide benefits. Compliance levels also vary across individuals

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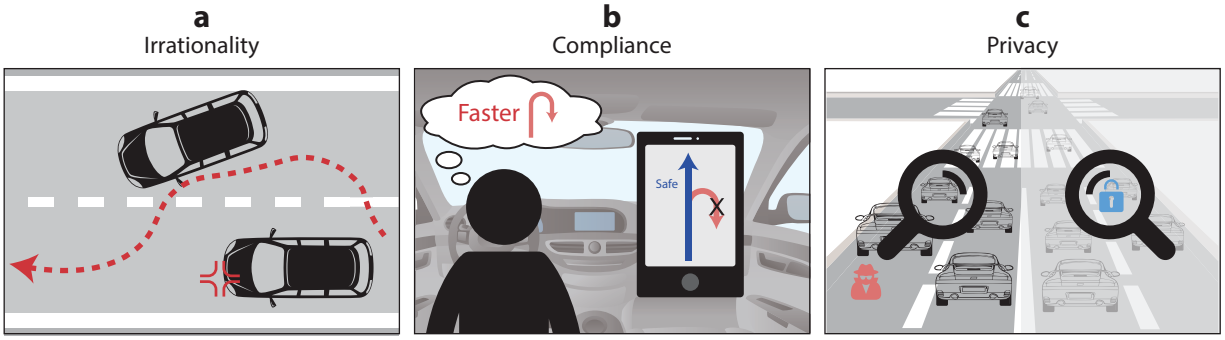


Figure 2

Challenges in mixed crowd navigation. (*a*) Irrationality: Humans deviate from rational behavior (e.g., traffic violations/unsafe driving actions). (*b*) Compliance: Robot decisions depend on human compliance, but humans may resist guidance or prioritizing goals. (*c*) Privacy: Sensitive individual data in crowds should be protected with anonymization and privacy-preserving techniques.

and contexts, necessitating methods to predict and incentivize cooperation, such as personalized rewards (150). Third, privacy and security remain critical concerns in mixed crowd perception and control. Systems must avoid leaking sensitive individual data, such as vehicle trajectories in traffic or pedestrian movement patterns, that could be exploited for malicious purposes. Recent work (139, 151) has proposed solutions such as federated learning and differential privacy to enable crowd-level learning and control without requiring granular individual data. Addressing these challenges will require interdisciplinary advances in behavioral modeling, incentive design, and privacy-aware algorithms.

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